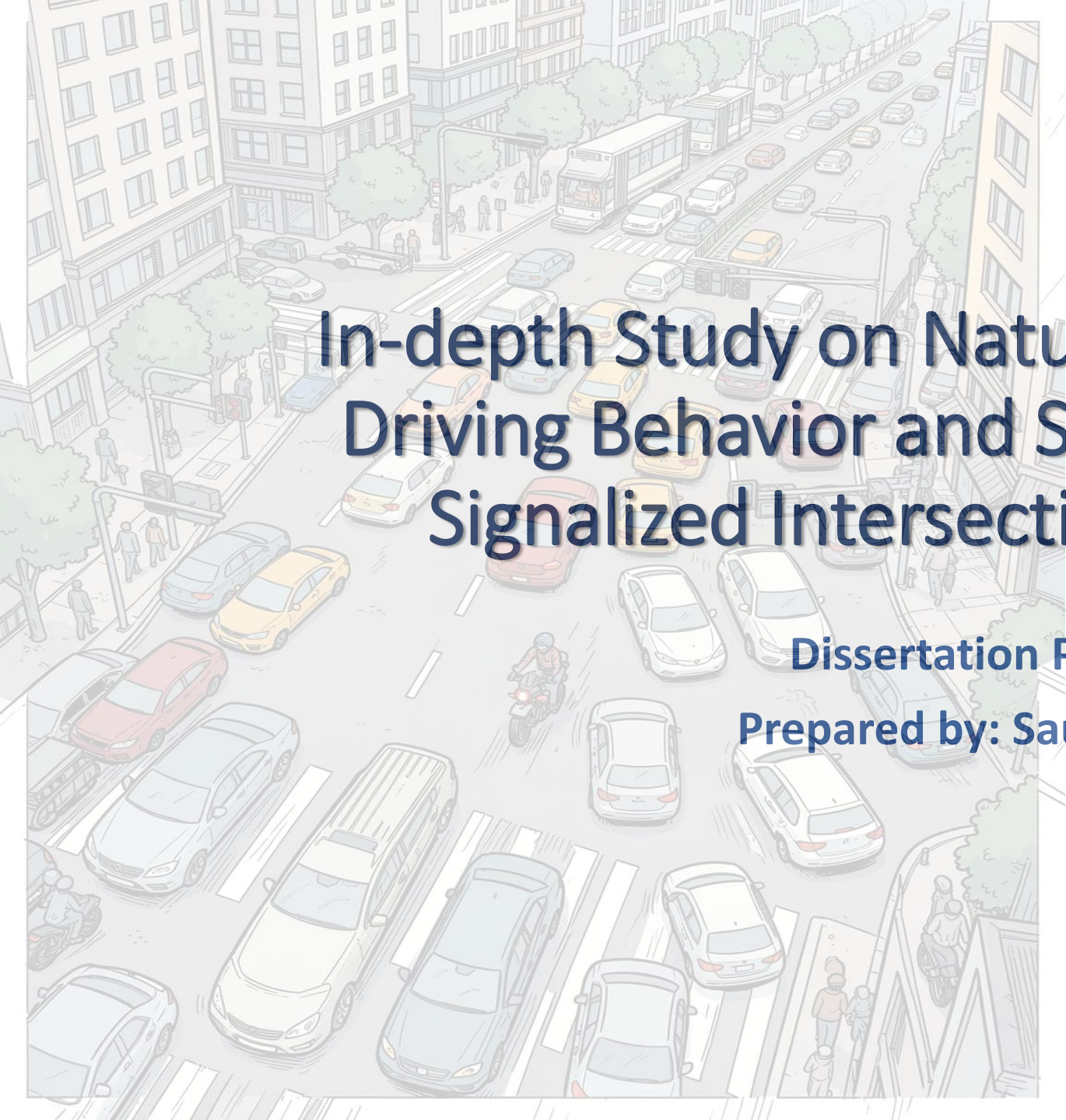


An illustration of a busy city intersection. The scene is viewed from an elevated perspective. In the center, a motorcycle is moving through the intersection. Surrounding it are various cars, including sedans and a van. A bus is visible on the left side of the intersection. Pedestrians are walking on the sidewalks. The background shows tall city buildings and trees lining the streets. The overall style is a detailed, colorful illustration.

In-depth Study on Naturalistic Motorcycle Driving Behavior and Surrogate Safety at Signalized Intersection Approaches

Dissertation Proposal

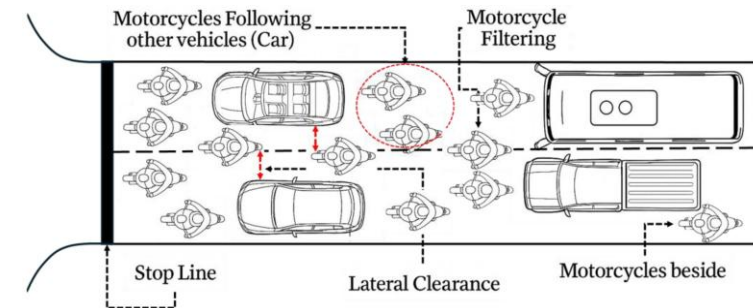
Prepared by: Saurav Barua

Introduction

- ❑ Motorcycles are a dominant mode of urban transportation in Bangkok, resulting in frequent interactions with surrounding vehicles at signalized intersections.
- ❑ Conventional traffic models often do not fully capture the **naturalistic and heterogeneous behavior of motorcycle riders**, particularly during queue formation and discharge.
- ❑ At signalized intersection approaches, motorcycles may exhibit distinctive behaviors such as **lane filtering, lateral maneuvering, queue jumping, and vehicle-type-specific following**.
- ❑ These behaviors are influenced by factors such as **lateral gap, speed differential, neighboring vehicles, vehicle type, and signal phase**.



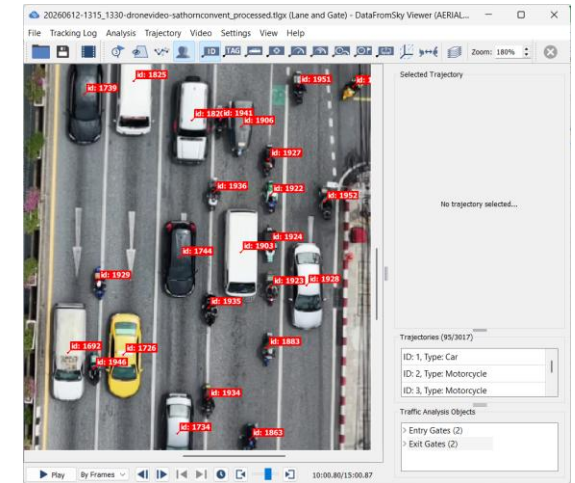
Motorcycle queue jumping at an intersection in Bangkok



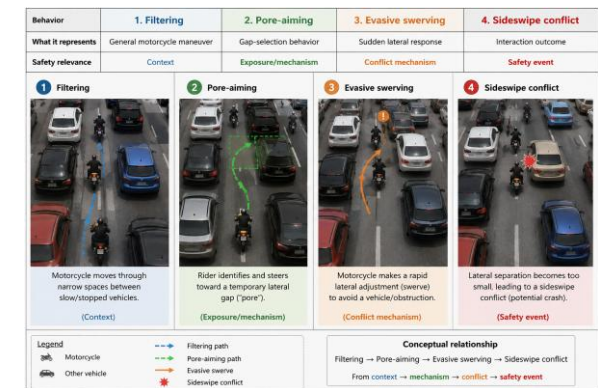
Conceptual illustration of motorcycle-vehicle interactions at a signalized intersection (Promraksa et al., 2022; Minh et al., 2012)

Introduction (Contd.)

- ❑ DataFromSky provides **high-resolution vehicle trajectory data**, enabling detailed observation of motorcycle–vehicle interactions and individual movement patterns.
- ❑ This study utilizes trajectory data collected at **different signalized intersections in Bangkok** to investigate the variability and consistency of naturalistic motorcycle driving behavior.
- ❑ The analysis focuses on **motorcycle interactions, maneuvering and lane filtering, surrogate safety, sub-lane behavior, and car-following behavior.**
- ❑ The findings will support the development of **behaviorally realistic motorcycle models and safety-oriented traffic simulation frameworks** for urban signalized intersections.



Top-down view of Motorcycles filtering in between cars in an intersection of Bangkok, screenshot from DataFromSky Viewer



Schematic illustration of different motorcycle maneuvering in a queue of a signalized approach

Contents

- Introduction
- Objectives
- Research Questions
- Terminologies and Definitions
- Literature Review
- Methodology

Research Objectives

Part A: Interaction, Maneuvering, Lane Filtering, and Safety

Objective 1

To characterize motorcycle–vehicle interaction and maneuvering at signalized approaches, and to identify the criteria governing lane filtering (lateral gap, speed differential, neighboring environment, adjacent vehicle type, signal phase).

Objective 2

To evaluate the safety of these interactions using surrogate safety measures (e.g., TTC, PET, DRAC, lateral-gap indicators) and to determine threshold values, safety index that identify unsafe motorcycle maneuvers.

Research Objectives (Contd.)

Part B: Simulation Calibration at Signalized Approaches

Objective 3

To calibrate a simulation model of motorcycle sub-lane behavior at signalized approaches, including lateral positioning and lane filtering, against observed trajectory data.

Objective 4

To estimate the proportion of motorcycles jumping to the front of the queue and to assess its effect on discharge flow (start-up lost time, saturation flow, headways) within the calibrated simulation.

Research Objectives (Contd.)

Part C: Car-Following Model Calibration

Objective 5

To calibrate motorcycle car-following models and develop regime-wise models (e.g., free-flow, following, closing, stop-and-go) that capture transitions in motorcycle following behavior.

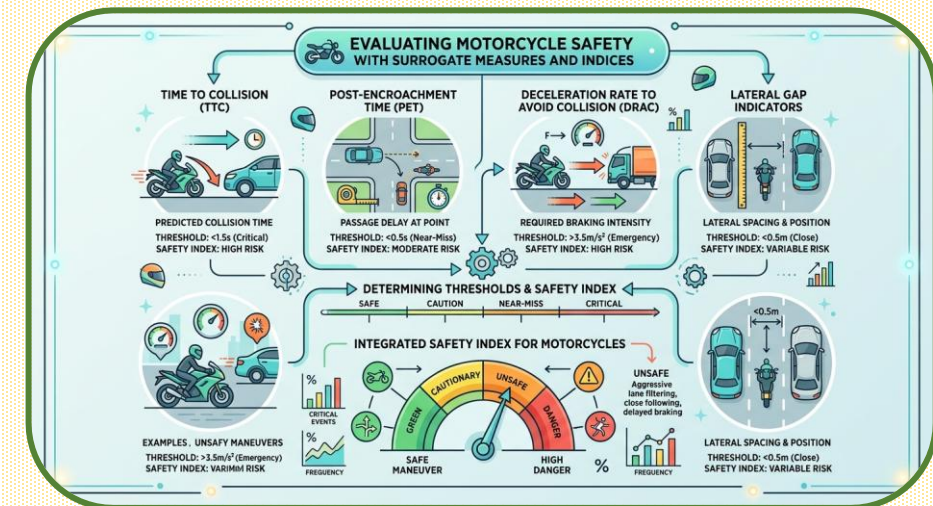
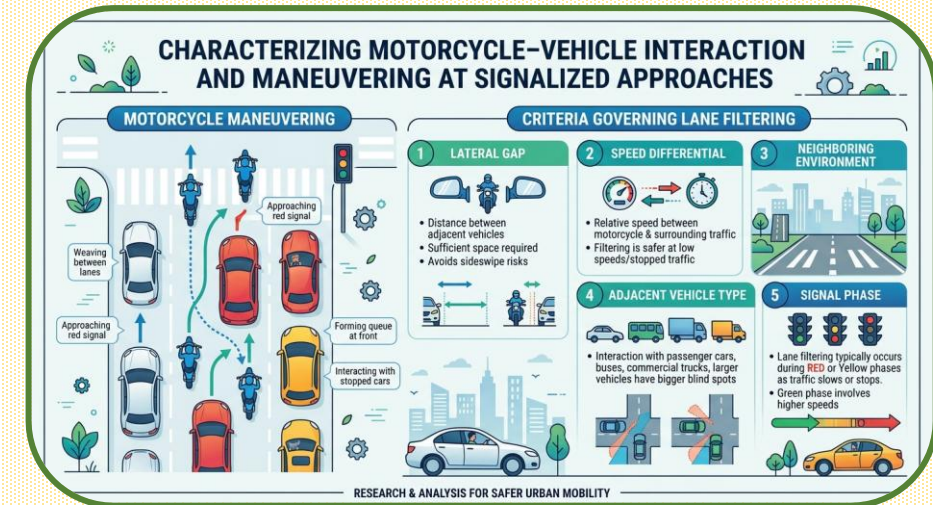
Objective 6

To calibrate vehicle-type-specific models for different leader–follower pairs involving motorcycles (motorcycle–motorcycle, motorcycle–car, motorcycle–heavy vehicle, car–motorcycle).

Research Questions

Part A: Interaction, Maneuvering, Lane Filtering, and Safety

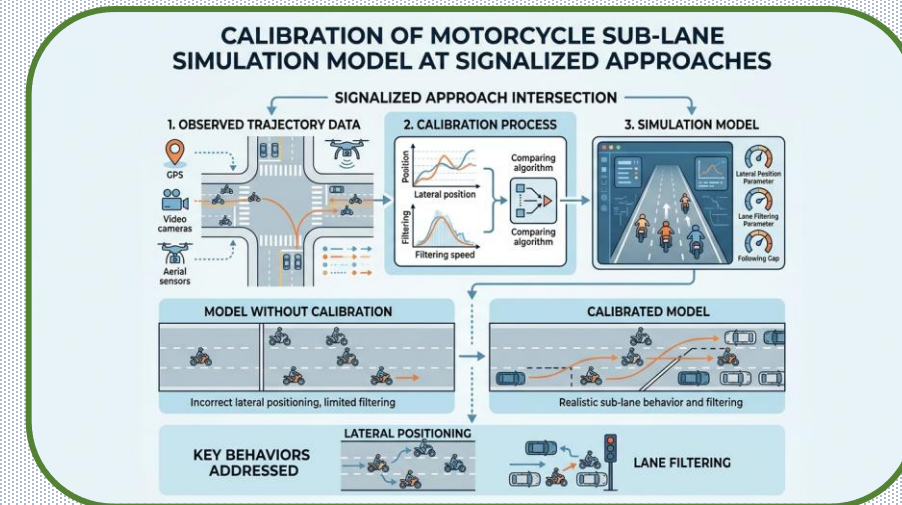
- **RQ1.** What conditions trigger motorcycle lane filtering, and what lateral gaps do riders accept when maneuvering past different vehicle types?
- **RQ2.** Which surrogate safety measures, and what threshold values, safety index best distinguish safe from unsafe motorcycle interactions during filtering and maneuvering?



Research Questions (Contd.)

Part B: Simulation Calibration at Signalized Approaches

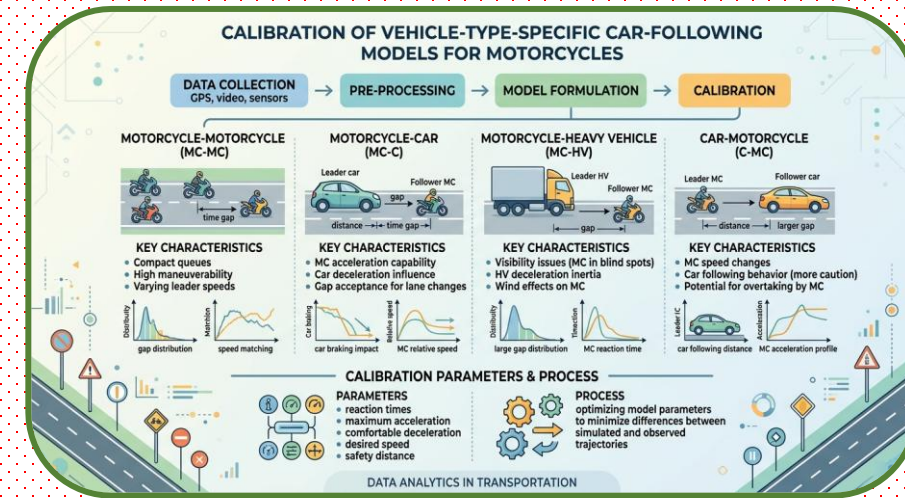
- **RQ3.** *Can a calibrated sub-lane model reproduce observed lateral positions and filtering behavior of motorcycles at the approaches of signalized intersections?*
- **RQ4.** *What proportion of motorcycles reach the start of the queue, and how does queue jumping alter discharge flow relative to simulation with default settings?*



Research Questions (Contd.)

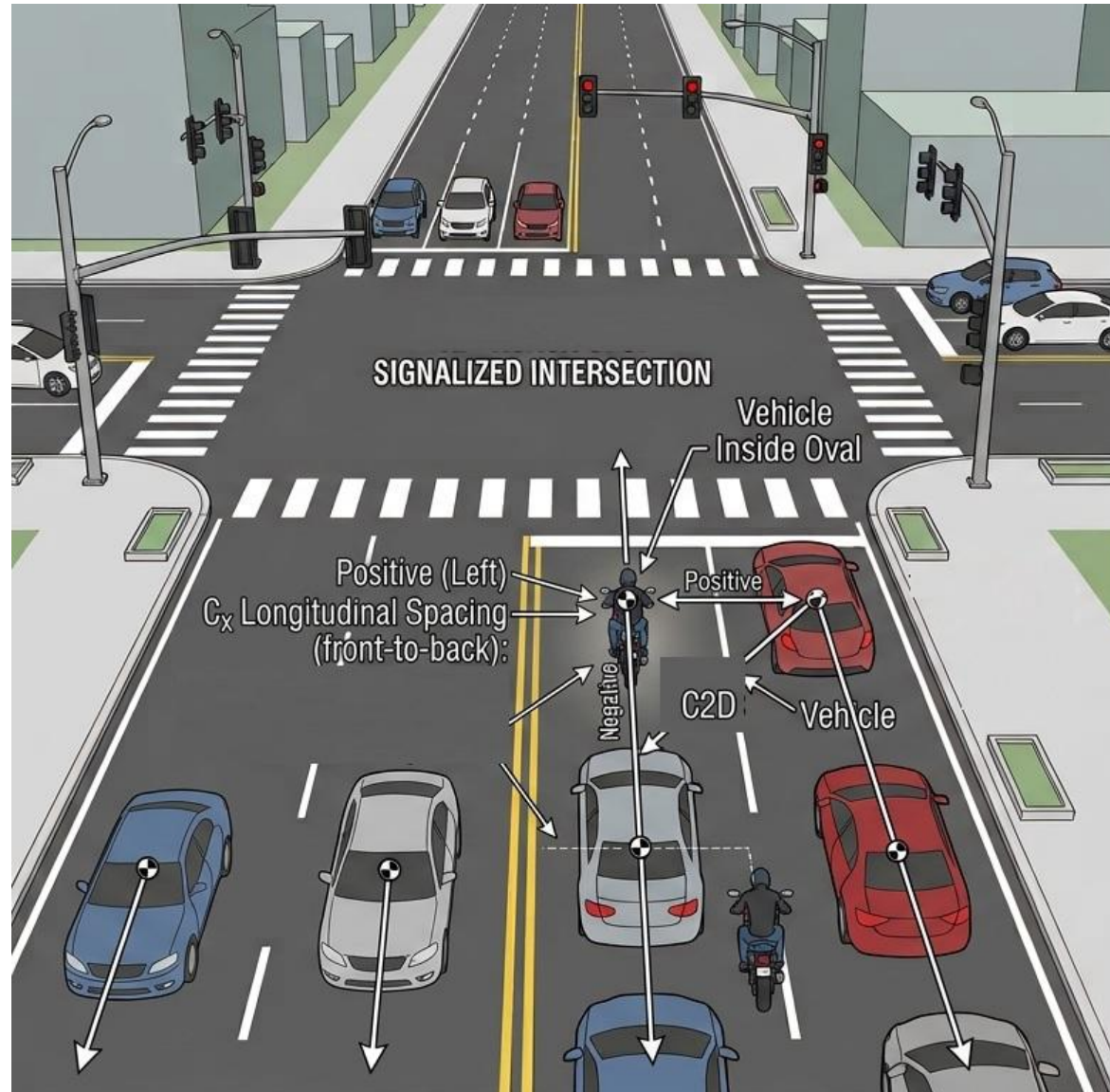
Part C: Car-Following Model Calibration

- **RQ5.** Do regime-wise car-following models reproduce observed motorcycle trajectories better than single-regime models, and what criteria define the regime transitions?
- **RQ6.** How do calibrated car-following parameters (reaction time, desired gap, sensitivity, acceleration) vary with the type of leading and following vehicle?



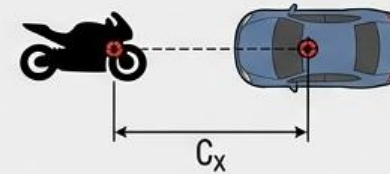
Terminologies and Definitions

Motorcycle 2D interactions



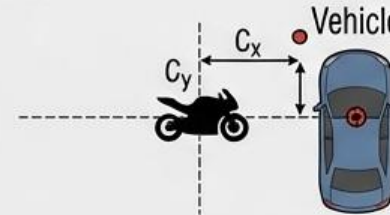
CONCEPTUAL EXPLANATION: VEHICLE SPACING MEASUREMENTS (CENTER-TO-CENTER)

C_x - Longitudinal spacing (front-to-back)



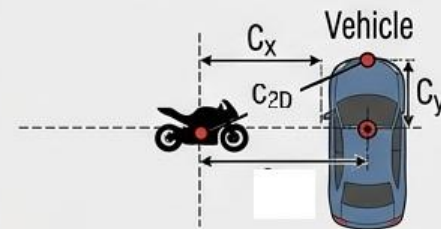
Measured along the direction of travel.
Positive C_x → other vehicle is ahead;
Negative C_x → other vehicle is behind.
(Think of it as gap in front/behind me)

C_y - Lateral spacing (side-to-side)



Measured perpendicular to the direction of travel.
Positive C_y → vehicle to your left;
Negative C_y → vehicle to your right.
(Think of it as room beside me)

C2D - Straight-line (2D) distance

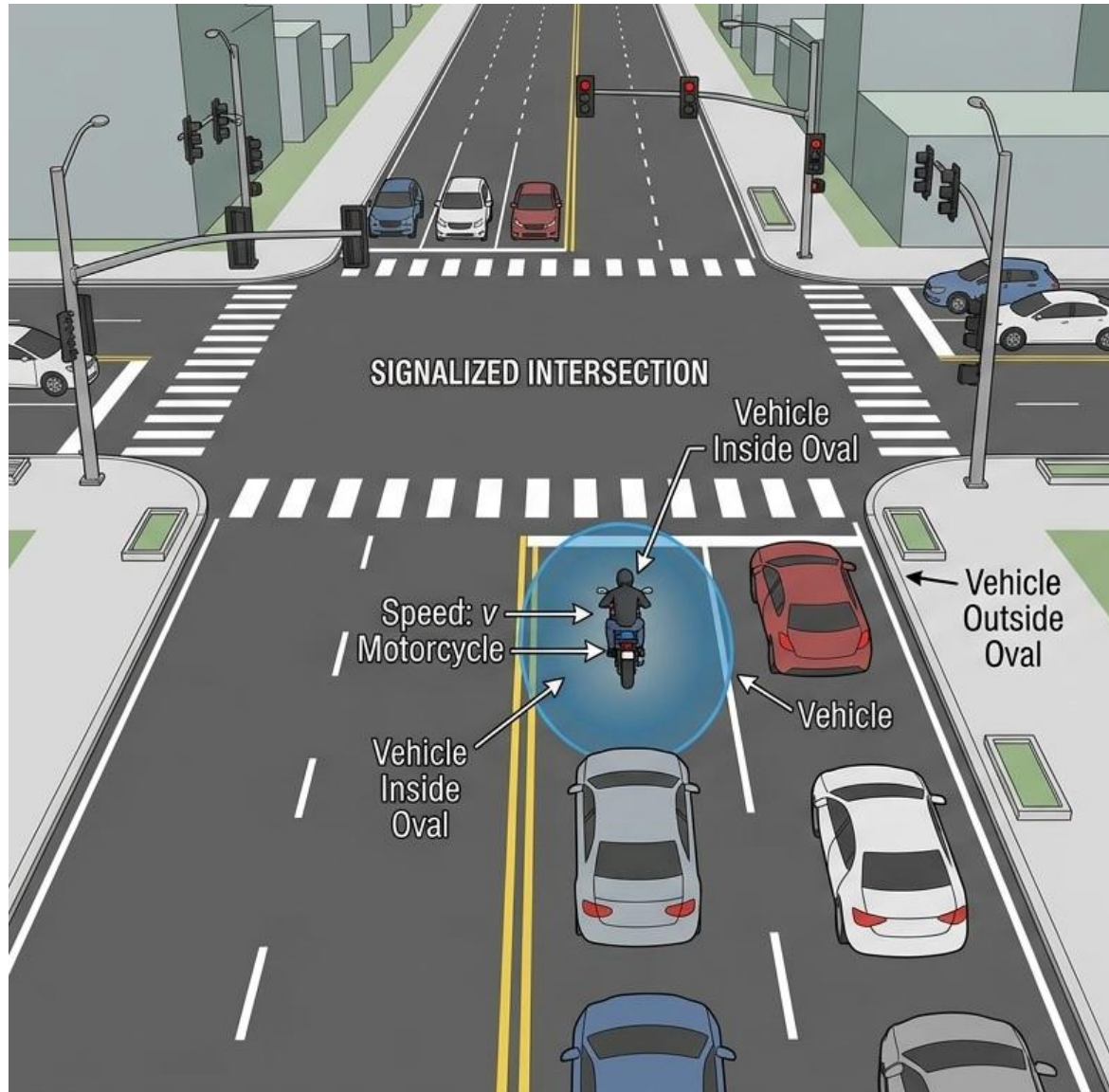


$$C_{2D} = \sqrt{C_x^2 + C_y^2}$$

Combines the two into one straight-line distance, like measuring "as the crow flies".

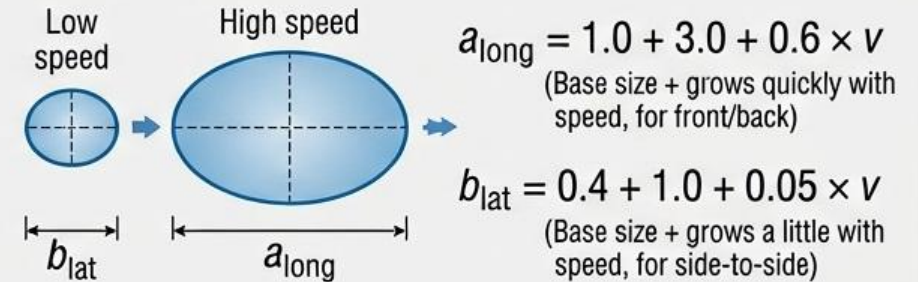
Terminologies and Definitions (Contd.)

Vehicle Envelop and Envelop index

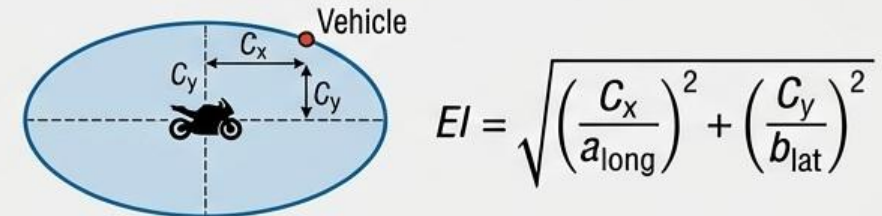


CONCEPTUAL EXPLANATION: THE MOTORCYCLE SAFETY OVAL

OVAL DIMENSIONS

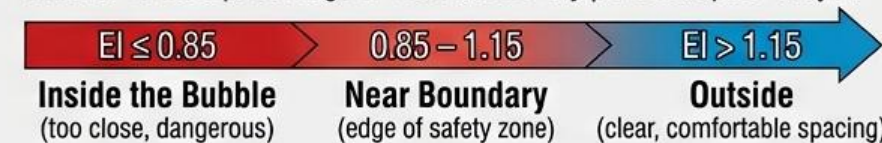


EI CALCULATION (ELLIPTICAL INDEX)



READING THE EI VALUE

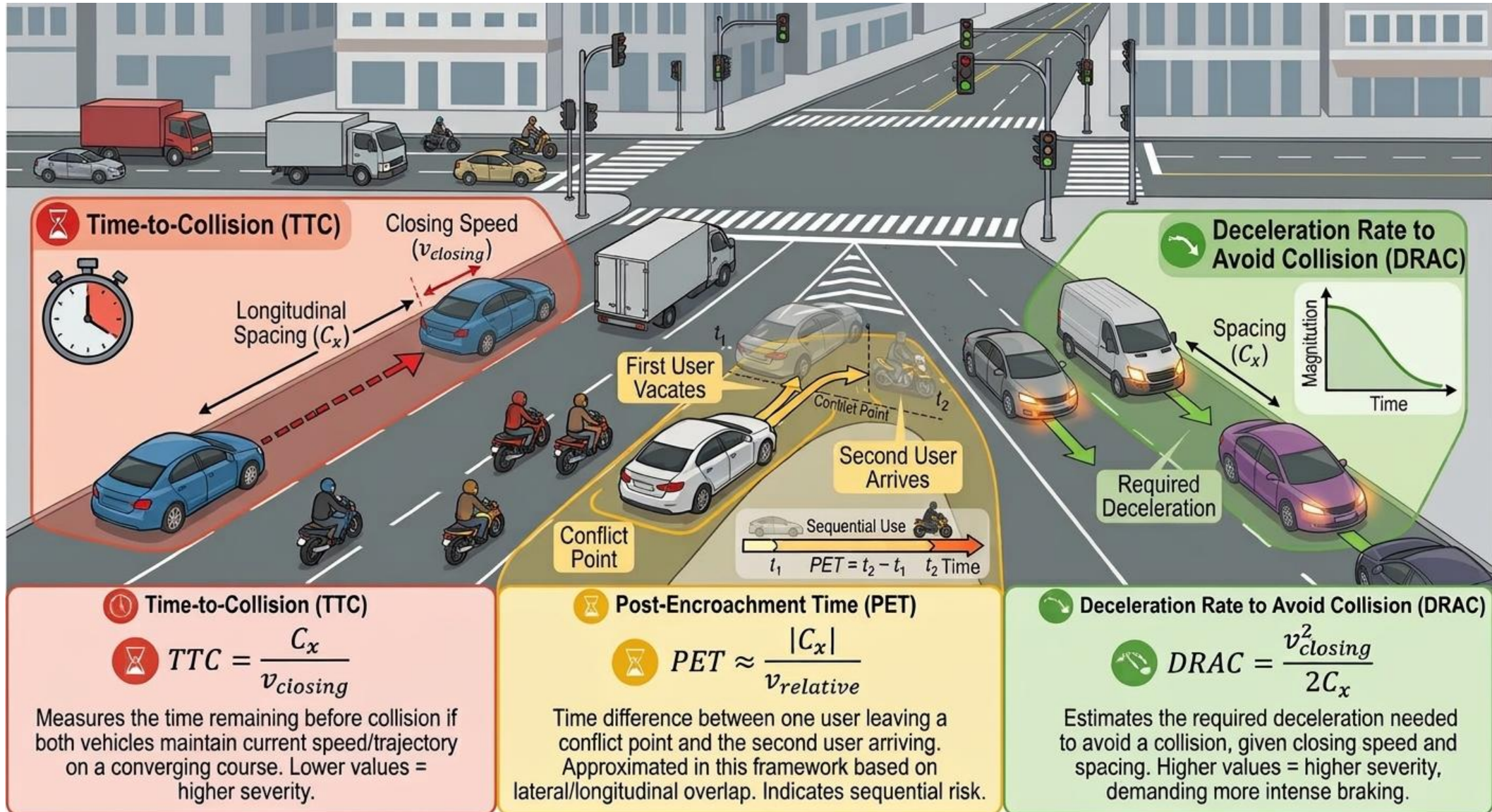
Think of EI like a percentage of "how far into my personal space are you"



Think of EI like a percentage of "how far into my personal space are you" – 0 means right on top of the motorcycle, 1.0 means exactly on the edge of the bubble, and anything above 1 means outside it.

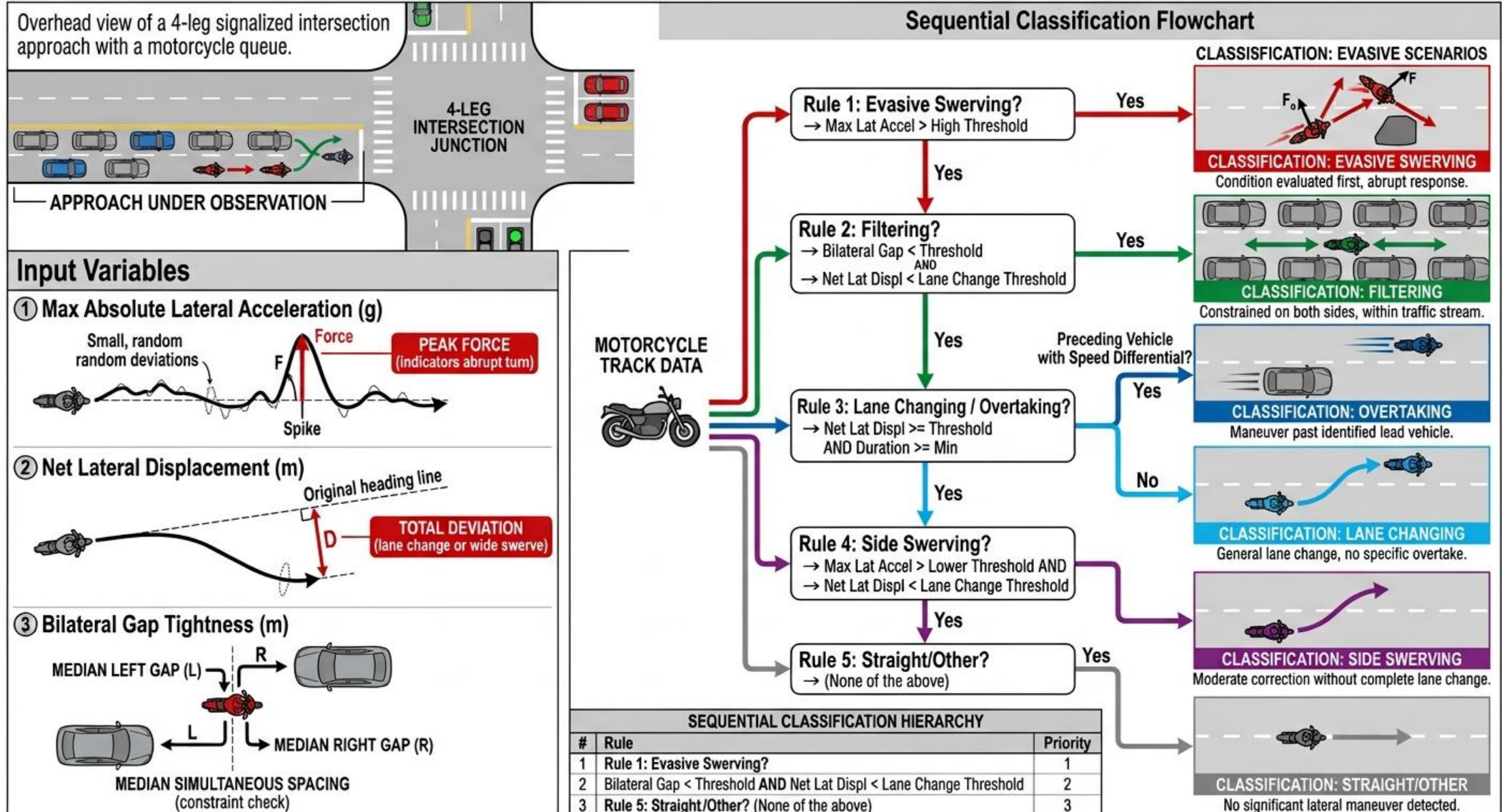
Terminologies and Definitions (Contd.)

Surrogate Safety Measures



Terminologies and Definitions (Contd.)

Motorcycle Maneuver Classification



Terminologies and Definitions (Contd.)

Motorcycle Maneuver Classification- Threshold

Threshold	Value	Meaning
EVASIVE_LAT_ACC_MS2	2.5 m/s ²	Peak absolute lateral acceleration above which a maneuver is flagged Evasive Swerving
SWERVE_LAT_ACC_MS2	1.5 m/s ²	Peak absolute lateral acceleration above which (but below the evasive threshold) a maneuver is flagged Side Swerving
FILTER_GAP_M	1.6 m	Max median lateral gap allowed on each side (left and right) for the "both sides tight" condition used in Filtering detection
LANE_CHANGE_LAT_DISP_M	2.2 m	Net lateral displacement (max – min of the cross-track offset) threshold; below this with tight gaps on both sides → Filtering; at/above this → candidate Lane Change/Overtake
LANE_CHANGE_MIN_DUR_S	1.0 s	Minimum track duration required (alongside the lateral displacement threshold) to qualify as a Lane Change/Overtake rather than noise
OVERTAKE_MIN_DSPEED_KMH	5.0 km/h	Minimum average speed advantage the motorcycle must have over an ahead-neighbor ($0 < C_x < 20$ m) to relabel a lane change as Overtaking rather than plain Lane Changing

Literature Review

Motorcycle–Vehicle Interaction

References	Methodology & Interaction Framework	Key Interaction Dynamics
Avathkattil, S., & Perumal, V. (2026)	2D spatial safety envelope extraction establishing 95%, 99%, and 100% physical clearance contours across vehicle classes and speed bins.	Mapped non-symmetric, speed-dependent physical safety envelopes , quantifying how lateral and longitudinal clearance margins expand around two-wheelers relative to cars.
Amini, E. et al.(2019)	High-resolution video tracking in Tehran; calibrated Gazis-Herman-Rothery (GHR) car-following model parameters via Genetic Algorithm.	Passenger car drivers exhibit heightened caution when following motorcycles , maintaining approximately 10 meters larger space headway at speeds <50 km/h than when following other cars.
Damani, J., & Vedagiri, P. (2024)	Video trajectory extraction (4,920 data points across 665 pairs); Generalized Linear Models (GLMs) comparing Strict Following (SF) vs. Oblique Following (OF) .	Distance gaps are smallest when following another motorcycle (4.87 m) and largest behind heavy vehicles (9.7 m). Oblique (staggered) following yields tighter gaps and higher conflict frequencies.
Torquato, R. et al. (2020).	Statistical regression on queue discharge headways evaluating motorcycle positioning (ahead, virtual side lane, longitudinal gaps) in Brazil.	Motorcycles positioned directly ahead of or between four-wheelers increased start-up lost time by 0.3 to 0.7 s, whereas motorcycles on virtual side lanes caused no queue delay.
Suweda, I. W., & Wedagama, D. M. P. (2021)	Empirical queue discharge headway timing across signalized approaches in Bali, Indonesia.	Motorcyclists shorten individual discharge headways due to rapid acceleration at green onset, but increase headways for adjacent four-wheelers due to lateral encroachment .

Literature Review (Contd.)

Lane Filtering and Maneuvering

References	Methodology & Data Source	Key Maneuvering Findings
Siridhara, T. et al.(2022)	Instrumented motorcycle equipped with laser sensors, GPS, and video logging lateral clearance choices in Bangkok queuing.	Identified riding speed, adjacent vehicle type (bus/truck vs. car), and physical gap width as the primary determinants governing motorcyclist lane-filtering choices .
Mulvihill, M. C. et al. (2013)	On-road verbal protocol analysis, Situation Awareness Global Assessment Technique (SAGAT), and helmet camera footage during live urban filtering.	Motorcyclists maintain high situational awareness during slow-speed lane filtering between stationary queues, actively scanning for opening vehicle doors and sudden lane changes.
Ouellet, J. V. (2012)	Caltrans freeway video sensor feeds combined with in-depth analysis of 900 motorcycle crash records from the USC Hurt Study.	Lane splitting drops sharply when traffic speeds exceed 40 mph. In heavy freeway congestion, lane-splitting riders are under-represented in crashes relative to traffic exposure (29% crash share vs. 63% exposure share).
Damani, J., & Vedagiri, P. (2020)	Comparative modeling using Binary Logit, Random Forest, and Adaptive Neuro-Fuzzy Inference Systems (ANFIS) in Indian traffic streams.	Clear lateral gap width and relative speed differences drive filtering decisions . ANFIS achieved superior predictive accuracy (>95%) over statistical logit models.
Samalla, S. et al.(2024)	Extreme Value Analysis (EVA) of surrogate crash events and high-resolution trajectory tracking in weak-lane-disciplined corridors.	Formulated non-stationary crash risk thresholds for powered two-wheelers during lateral filtering and inter-vehicle squeezing maneuvers between heavy vehicles.

Literature Review (Contd.)

Motorcycle Driving Behavior

References	Methodology & Data Source	Key Behavioral Findings
Horswill, M. S., & Helman, S. (2003)	Laboratory video simulation tests and unobtrusive roadside field observations comparing matched driver and rider cohorts.	Motorcyclists chose significantly faster travel speeds, engaged in more frequent overtaking, and accepted smaller gaps than non-riding car drivers. However, their superior hazard perception skills persisted even when operating passenger cars.
Jensupakarn, A., & Kanitpong, K. (2018)	Observational field survey and binary logistic regression modeling comparing 1,200 motorcyclists and car drivers in Thailand.	Motorcyclists demonstrated a significantly higher probability of red light running (RLR) than car drivers, primarily driven by front-of-queue positioning, younger rider age, and lack of automated enforcement.
Abdulwahid, S. N. Et al. (2022)	Machine learning classification (SVM, Random Forest) applied to real-time telemetry (heart rate, speed, acceleration, lean angle).	Achieved >93% accuracy in classifying aggressive riding behaviors (speeding, weaving, tailgating) , explicitly linking rider physiological stress responses to dangerous traffic interactions.
Halim, W. et al.(2025)	Naturalistic driving study (NDS) tracking 79 motorcyclists across 6 Indonesian cities covering 1,036 km of real-world riding.	Identified primary safety-critical event triggers: human error, overtaking against opposing traffic flow (18.9%–29.5%), unhelmeted riding, and sudden un-signaled lane departures.

Literature Review (Contd.)

Surrogate Safety Measures (SSMs)

References	Methodology & Analytical Approach	Key Safety & Conflict Insights
Wang, C. et al. (2021)	Systematic literature review of temporal, distance, and deceleration SSMs (TTC, PET, DRAC, CPI) and microscopic simulation models.	Synthesized time-based and deceleration-based SSM formulations , detailing calibration needs for mixed traffic streams and CAV platoon safety.
Borsos, A. (2021)	Bivariate threshold excess models (Peak Over Threshold using logistic distributions) and copula analysis on intersection trajectories.	Proved that temporal proximity (TTC) and speed indicators (relative speed, Delta-V) exhibit weak asymptotic dependence , demonstrating that time-based indicators alone cannot assess crash severity.
Maji, A., & Ghosh, I. (2025)	High-resolution drone video, SSAM conflict analysis, Peak over Threshold (POT) bivariate EVT , AdaBoost, and Ordered Probit models.	Determined critical safety conflict thresholds for TTC (1.25 s) and MaxDeltaV (4.5 m/s) , showing circular roadway geometry directly dictates conflict frequency.
Wu, Y. et al. (2020)	ARCIS framework: Mask R-CNN detection, CSRT tracking , and pixel-level PET calculation from aerial UAV video.	Automated extraction of pixel-level PET values, achieving high accuracy in classifying conflict types (rear-end, sideswipe, angle, head-on) at signalized junctions .
Lu, C. et al. (2021)	Evaluated 6 longitudinal SSMs (TTC, S-PDRF, DRAC, PICUD, PFS, CFS) using 100-Car Naturalistic Driving Study dataset.	Concluded S-PDRF is most robust against sensor noise , whereas TTC is the most computationally efficient indicator for real-time safety systems.

Literature Review (Contd.)

Motorcycle Modeling and Simulation

Reference (APA Style)	Methodology & Modeling Architecture	Key Simulation Findings
Minh, C. C. et al.(2007)	Formulated the dynamic motorcycle lane concept ($lw = 0.07V + 0.80$) and perceptual threshold distance (S_n); calibrated via maximum likelihood.	Reaction times follow a log-normal distribution (mean 0.79 s, SD 0.24 s). Rider gender and passenger occupancy significantly influence acceleration/deceleration rates.
Hsu, T.-P., & Chen, Y. (2012)	Synthesizes gravitational car-following with Helbing's 2D social force model (attractive forces to open space, repulsive forces from obstacles).	Successfully replicated continuous 2D lateral swerving and longitudinal filtering kinetics . Achieved a speed RMSE of 2.78 m/s and direction angle MAPE of 3.0%.
Trinh, L. T. et al.(2018)	Combines anticipation movement logic with 2-player game theory (payoffs based on travel time through conflict zones) in NetLogo/Java.	Replicated >70% of actual left-turning motorcycle trajectory shapes observed in field datasets, capturing non-lane conflict resolutions ('cut tail' vs. 'giving way') .
Lee, T.-C. et al. (2009)	Agent-based simulator (BikeSim) incorporating 8 motorcycle movement patterns , MNL path choice ($p^2 = 0.47$), and Gipps car-following in Java.	Replaced rigid lane adherence with dynamic virtual lanes , accurately reproducing London's mixed-traffic spatial distribution, filtering, and queue discharge capacity.
Wichitphongsa, W. et al. (2025)	Customized External Driver Model (EDM) embedded in PTV Vissim 2024 to simulate motorcycle box intersections (advanced stop lines).	Demonstrated that motorcycle box intersections increase lane-filtering rates by +26.7% and reduce motorcycle departure delay without impacting passenger car queues.

Methodology

Motorcycle Traffic Modeling and Safety Analysis

1. Data Collection and Extraction

Collect drone video at signalized intersections in Bangkok and extract trajectories using DataFromSky.

3. Interaction and Filtering Analysis

Identify motorcycle-vehicle interactions, develop empirical and speed-dependent interaction envelopes, and determine lane-filtering criteria.

5. Sub-Lane Model Calibration

Calibrate and validate a motorcycle sub-lane/agent-based simulation model using observed lateral movement, filtering, gap acceptance, speed, and discharge behavior.

7. Regime-Wise Modeling

Identify free-flow, following, closing, and stop-and-go regimes, calibrate models, and develop a regime-switching model.



2. Trajectory Processing and Validation

Clean, smooth, georeference, and validate trajectories; derive position, speed, acceleration, heading, and signal-phase information.

4. Safety Evaluation Using SSMs

Calculate TTC, PET, DRAC, lateral gap, and related 2D safety measures; develop safety thresholds and identify risky conditions.

6. Queue-Jumping and Impact Analysis

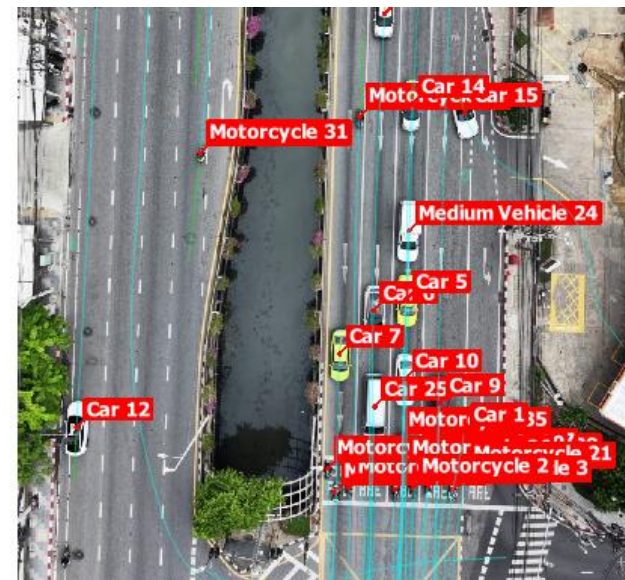
Quantify motorcycle queue-jumping behavior and evaluate effects on start-up lost time, headway, and saturation flow.

8. Vehicle-Type-Specific Model Development

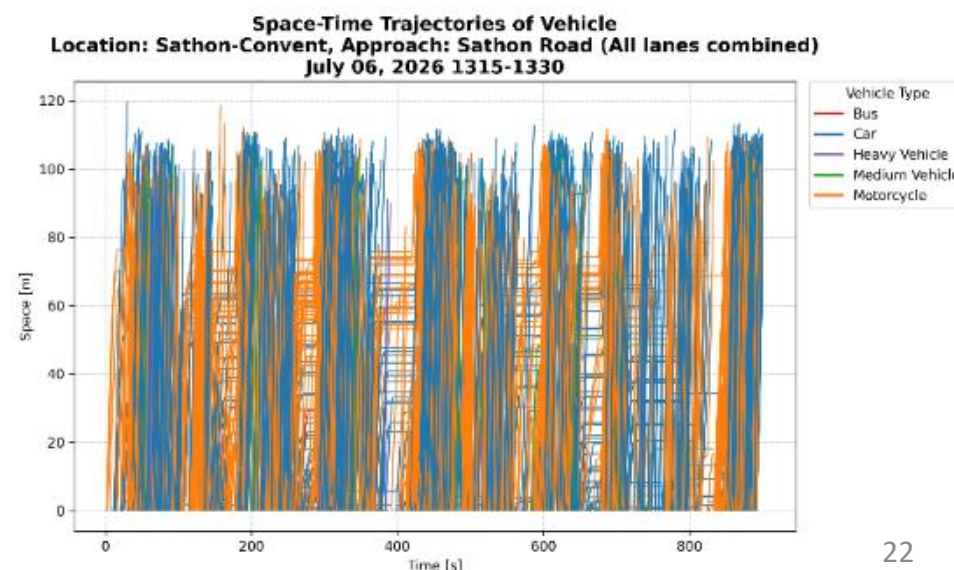
Calibrate and compare models for MC-MC, MC-Car, MC-HV, and Car-MC combinations and validate transferability.

Data collection & Extraction

- ✓ Collect aerial drone video covering the approach area of the selected signalized intersections in Bangkok.
- ✓ Process videos using **DataFromSky**.
- ✓ Extract and classify vehicle trajectories.
- ✓ Obtain **ID, type, x-y position, time, speed, and acceleration**.
- ✓ Convert trajectories to real-world coordinates and smooth data.
- ✓ Validate and filter trajectory quality.



Screenshot from Drone video footage of traffic



Space-time vehicle trajectory diagram

Trajectory Data Processing Workflow

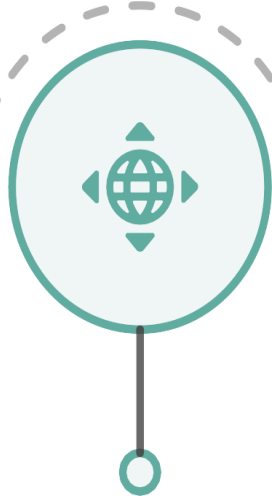
1. Clean and filter data

Clean and filter incomplete or erroneous trajectories.



2. Convert coordinates

Convert coordinates to real-world metric coordinates.



3. Smooth and denoise

Smooth trajectory data and remove noise.



4. Derive motion metrics

Derive speed, acceleration, and heading.



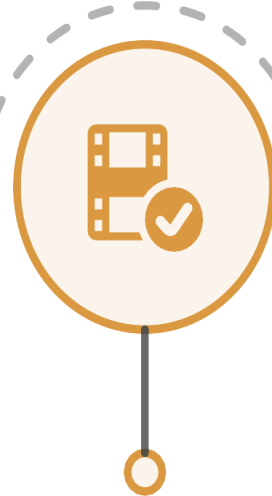
5. Check tracking continuity

Check tracking continuity and ID switches.



6. Validate trajectories

Validate selected trajectories against video observations.

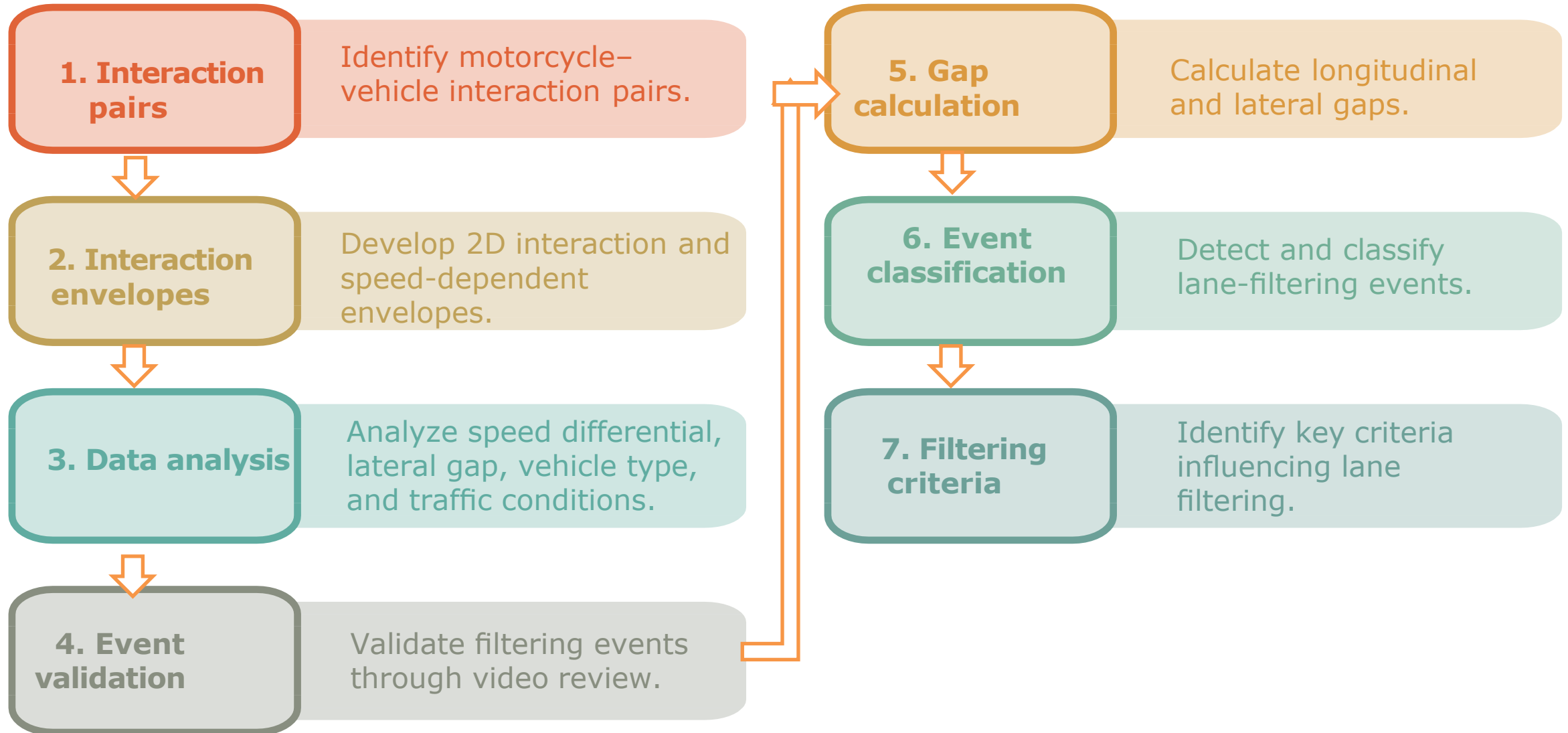


7. Calculate errors

Calculate position and speed errors.



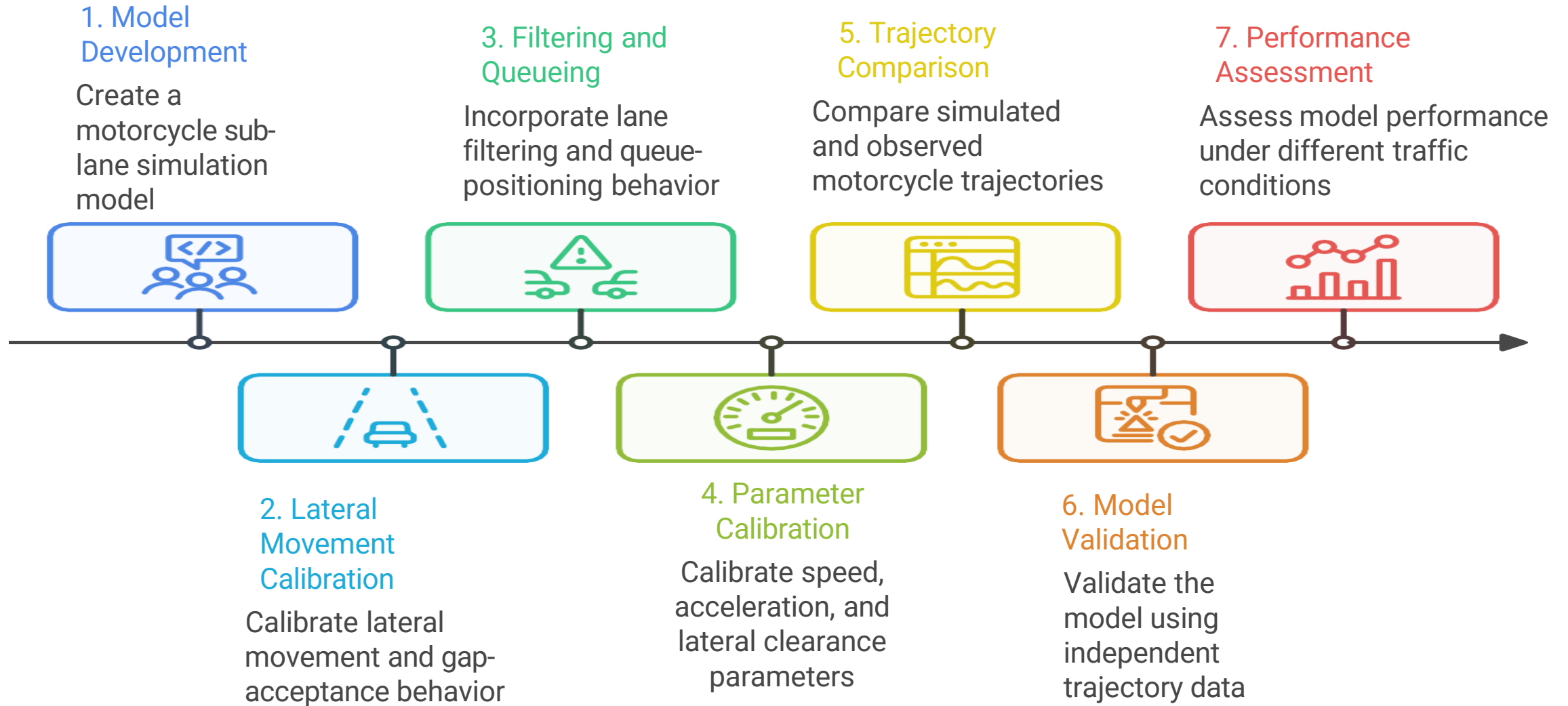
Motorcycle Interaction & Lane-Filtering Analysis



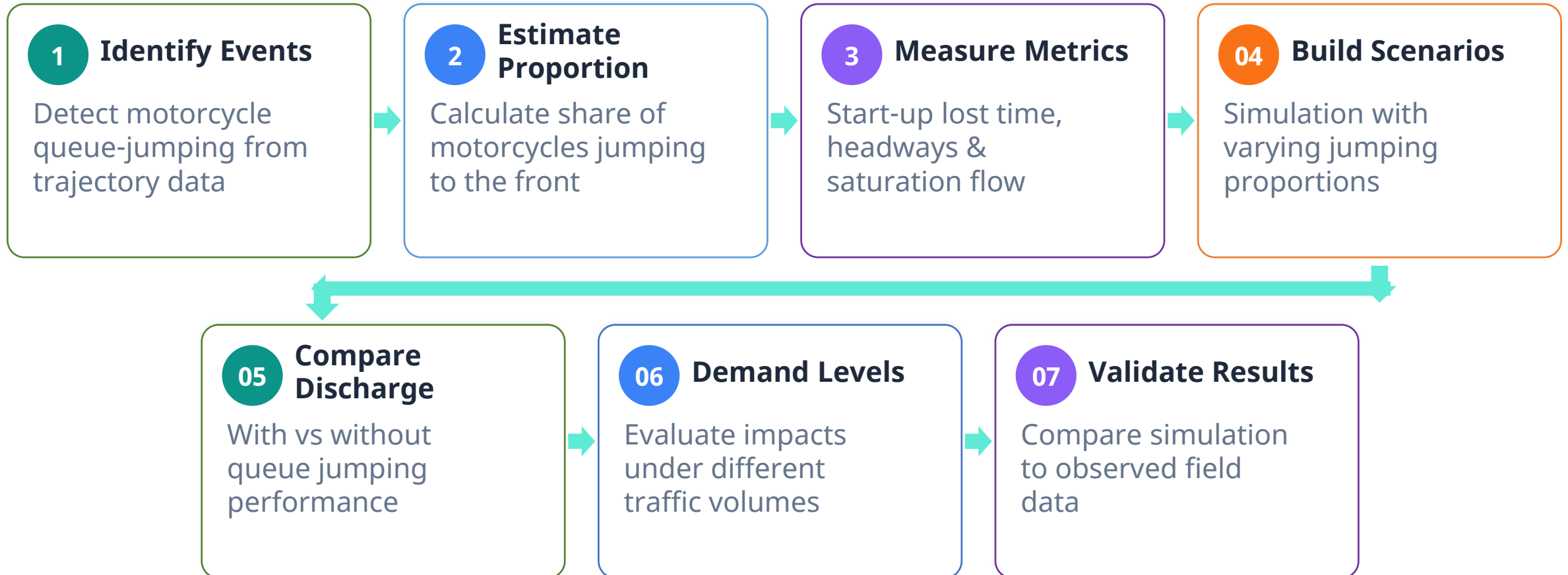
Safety Evaluation Using SSMs



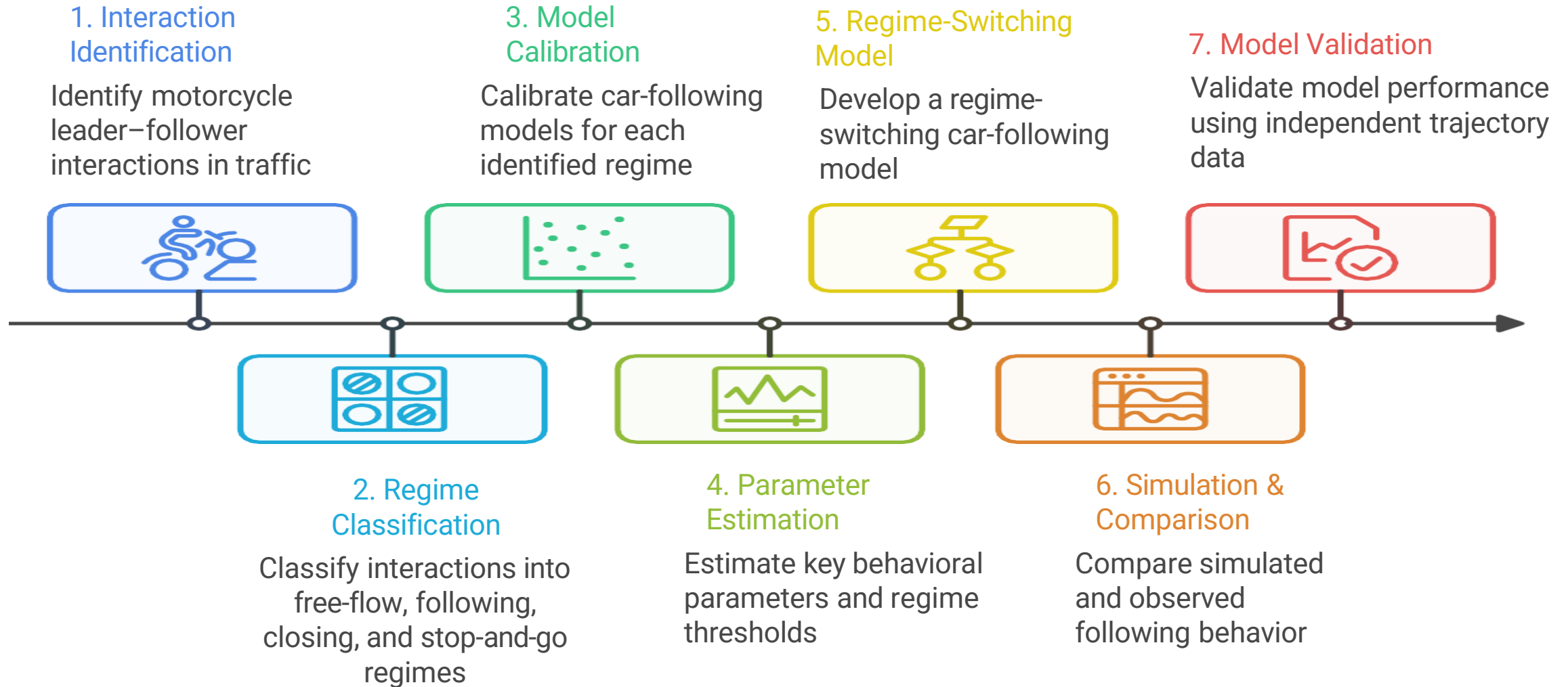
Motorcycle Sub-Lane Simulation Model



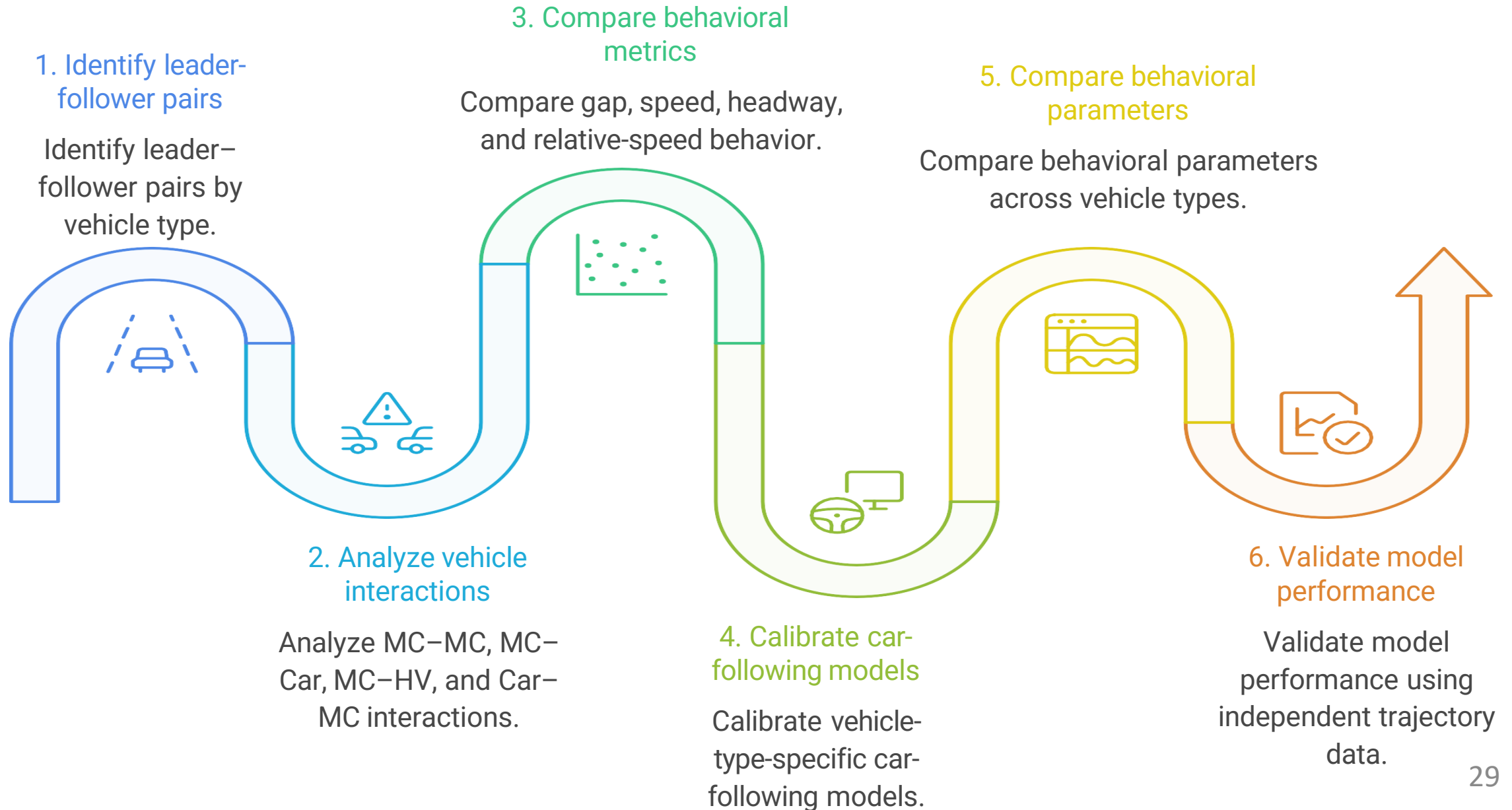
Queue-Jumping & Discharge Impact Analysis



Motorcycle Car-Following Model Calibration



Vehicle-Type-Specific Car-Following Model Calibration



Thank You